Working with AI

Lessons from `CountChar`

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COSC 2230 · CS III · Augustana University

Fall 2026

You used AI - and you said so

82%

of you used AI for `CountChar`

- About the same share as professional developers
- Saying so is right. The question is **how** you use it.

Developers: JetBrains State of Developer Ecosystem 2025 - 85% of 24,534 developers use AI regularly.

Good uses this time

- A checklist of everything the assignment asks for
- An explanation of a compiler error
  `no suitable method found for count(String,char,int,int)`
- An explanation of a concept: why Java does not reuse the frame of a last call
- A check of a trace you had already written
- Formatting a table after you had worked out the values

Bad uses

- The AI writes the methods, the trace and the answers - you hand them in
- Pasting “fixed” code without running it
- Asking “is my solution correct?” instead of running it
- Rewriting the program with other libraries because `stdlib.jar` did not work

You learn nothing

The in-class checks and the exams will show it.

Habits that make the difference (1)

1. **Run it yourself before you hand it in.**
   Compile, run, compare with the sample output line by line. "The AI says it is correct" is not a test.
2. **Follow the assignment, not the AI.**
   AI tools add imports (`algs4`), rename classes, "improve" given code.
3. **Ask why before you ask for a fix.**
   Whole file + exact error message: what causes it?

Habits that make the difference (2)

4. **Do the thinking parts by hand first.**
   Write the trace, then ask the AI to check it.
   Most common error: `countAcc("banana", 'a', 1, 1)` instead of `(1, 0)`
5. **Answer every part of the question.**
   Ask the AI: "Which parts of the question does my answer not cover?"
6. **Fix your setup once.**
   `javac` or `stdlib.jar` does not work? Come to office hours.

Habits that make the difference (3)

7. **Make your AI statement specific.**
   At the top of the file: which tool, for which part, what you checked yourself - or "none".

From now on

Graded: tool, part, what you checked - not how much you used.
One part missing: -0.5 points. No statement: -1 point.

Example:

```
AI use: ChatGPT (tool) to explain the compiler error in countLoop and
to check my trace in Q1 (part). I wrote the trace first and ran the
program against the sample output myself (what I checked).
```

What comes next

- AI challenges all of us - a lasting change, not a passing trend
- My job: prepare you for your professional future
- Course structure, content, sources, delivery, assessment: all under review
- Augie CS club: a session on the future of software engineering
  (this term or next)